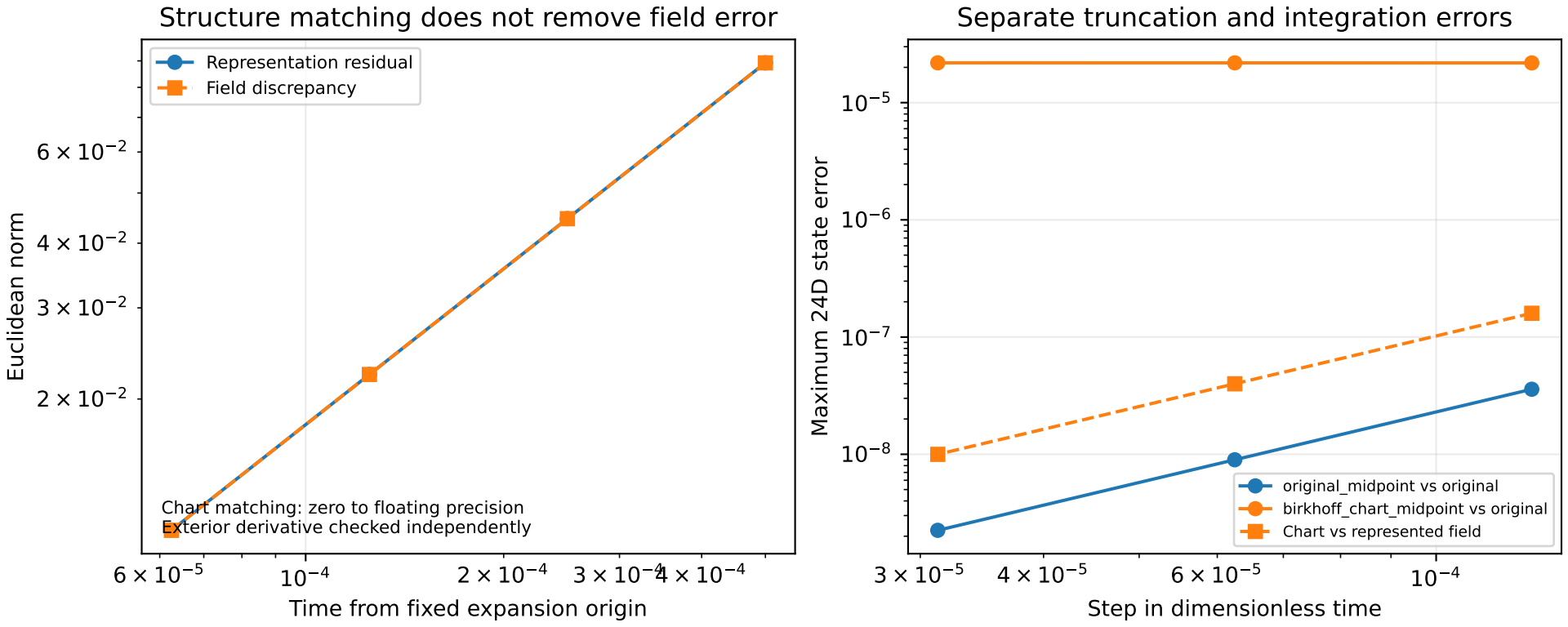
Structure matching does not remove field error
Representation residual
Field discrepancy
Euclidean norm
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
Chart matching: zero to floating precision
Exterior derivative checked independently
6×10^{-5}
10^{-4}
2×10^{-4}
3×10^{-4}
4×10^{-4}
Time from fixed expansion origin
Separate truncation and integration errors
Maximum 24D state error
10^{-5}
10^{-6}
10^{-7}
10^{-8}
original_midpoint vs original
birkhoff_chart_midpoint vs original
Chart vs represented field
3×10^{-5}
4×10^{-5}
6×10^{-5}
10^{-4}
Step in dimensionless time